

Trainings ▾

General Information

Connecting Rods

This familiarization section helps to identify and orient the various styles of ISB connecting rods for installation. These details are important because some connecting rods are **not** marked with a part number and identification can be difficult. This section will also discuss the acceptability of mixing connecting rods with different part numbers to make a repair.

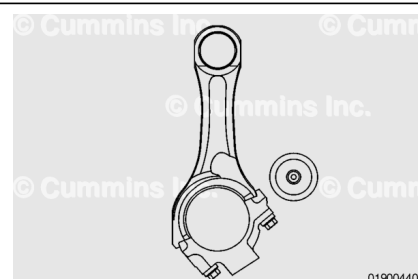
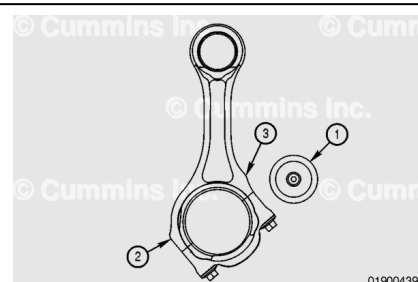
B Series engines use various connecting rods installed in different orientations, depending on where and when the engine was built. Generally, connecting rods should be installed in the same orientation as they were prior to removal. In special circumstances, this familiarization section may be referenced as a guideline.

The following information illustrates connecting rod styles and orientations for the Cummins® B Series 3.9 Liter (4 cylinder) and 5.9 Liter (6 cylinder) engines.

Each illustration shows the connecting rod and camshaft (1) as viewed from the front of the engine. The connecting rod capscrew holes help define the orientation of the connecting rod to the camshaft. One side of the rod has a through hole (2) and the other has a blind hole (3). The through hole is drilled through the rod. The blind hole is drilled into the rod and does not have an exit hole.

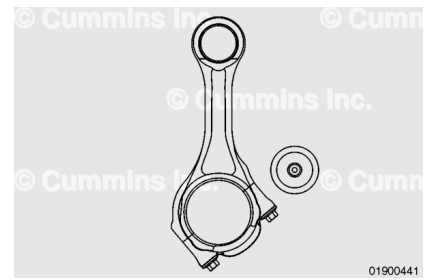
There are two possible orientations for the connecting rod:

- Through hole (2) toward the camshaft
- Blind hole (3) toward the camshaft.



Connecting Rod Part Number:	3942581 (3942581 shown)
Cylinder Block:	ISB CM550, ISB CM850
Connecting Rod Joint:	Machined
Cylinder Bore:	102 mm [4.015 in]
Cylinder Stroke:	120 mm [4.724 in]
Orientation:	Blind hole toward camshaft

Connecting Rod Part Number:	3954658, 3971212, 4891176, 4935880, 4943979, 4989163, 5257364 (3954658 shown)
Cylinder Block:	ISB CM550
Connecting Rod Joint:	Fractured
Cylinder Bore:	102 mm [4.015 in]
Cylinder Stroke:	120 mm [4.724 in]
Orientation:	Blind hole toward camshaft



Connecting Rod Part Number:	3954658, 3971212, 4891176, 4935880, 4943979, 4989163, 5257364 (3954658 shown)
Cylinder Block:	ISB CM850
Connecting Rod Joint:	Fractured
Cylinder Bore:	102 mm [4.015 in]
Cylinder Stroke:	120 mm [4.724 in]
Orientation:	Through hole toward camshaft

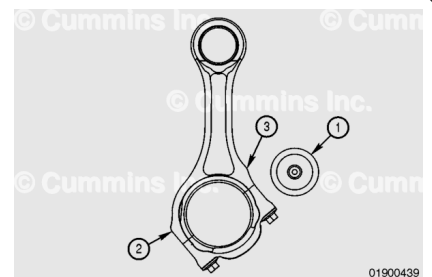


The following information illustrates connecting rod styles and orientations for the Cummins® B Series 4.5 Liter (4 cylinder) and 6.7 Liter (6 cylinder) engines.

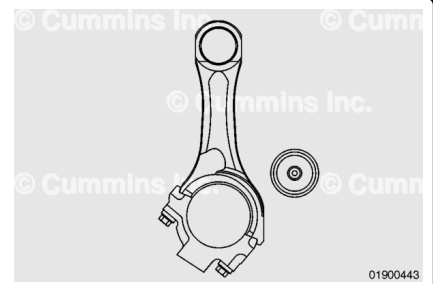
Each illustration shows the connecting rod and camshaft (1) as viewed from the front of the engine. The connecting rod capscrew holes help define the orientation of the connecting rod to the camshaft. One side of the rod has a through hole (2) and the other has a blind hole (3). The through hole is drilled through the rod. The blind hole is drilled into the rod and does not have an exit hole.

There are two possible orientations for the connecting rod:

- Through hole (2) toward the camshaft
- Blind hole (3) toward the camshaft.



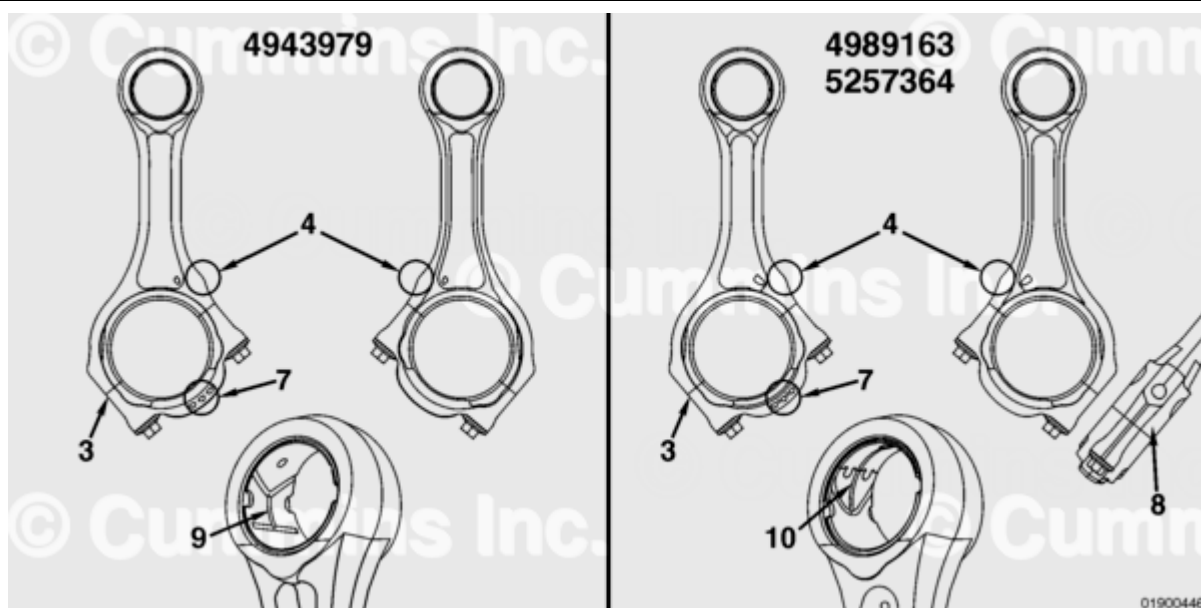
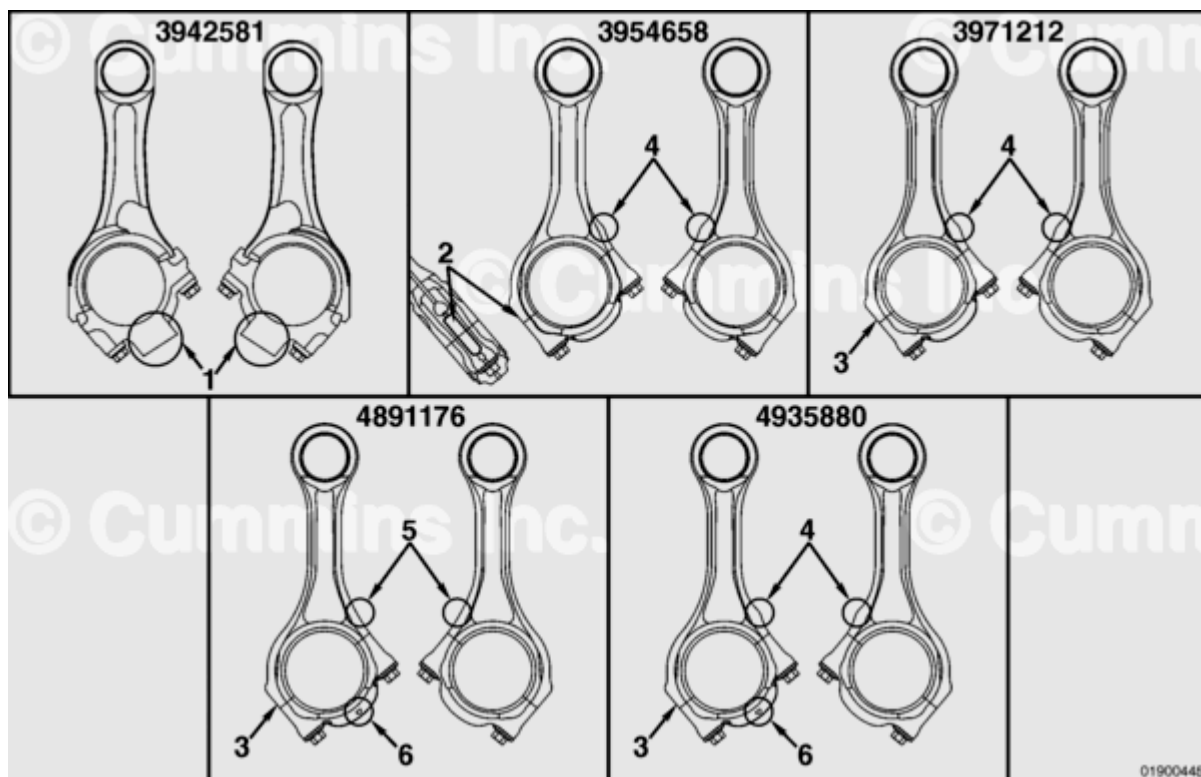
Connecting Rod Part Number:	3942581 (3942581 shown)
Cylinder Block:	ISB 4.5, ISB 6.7
Connecting Rod Joint:	Machined
Cylinder Bore:	107 mm [4.212 in]
Cylinder Stroke:	124 mm [4.881 in]
Orientation:	Through hole toward camshaft



Connecting Rod Part Number:	3954658, 3971212, 4891176, 4935880, 4943979, 4989163, 5257364 (4943979 shown)
Cylinder Block:	ISB 4.5, ISB 6.7
Connecting Rod Joint:	Fractured
Cylinder Bore:	107 mm [4.212 in]
Cylinder Stroke:	124 mm [4.881 in]
Orientation:	Blind hole toward camshaft



Use the following illustrations to identify 3.9 Liter, 4.5 Liter, 5.9 Liter, and 6.7 Liter (6 cylinder) connecting rods.



Note: Lead-free bushings (10) Part Numbers 4983518 and 5257363 are not serviceable.

Connecting Rod Characteristics

1. Balancer
2. Machined surface
3. Non-machined surface
4. Bump or bulge
5. Smooth
6. Single dimple

7. Three dimple
8. "TA LF" or "7364"
9. Oil grooves in leaded bushing Part Number 4891178
10. Oil grooves in lead-free bushings Part Numbers 4983518/5257363. (These bushings are **not** serviceable)

Due to weight differences that will result in engine vibration, special consideration **must** be given to mixing certain connecting rods in a single engine.

- In engines built before January 1, 2010, with connecting rod Part Numbers 4935880, 4943979, 4989163, and 5257364: these rods may **only** be mixed with each other.
- In engines built after January 1, 2010, with connecting rod Part Numbers 4989163, and 5257364: these rods may **only** be mixed with each other.
- Connecting rod Part Number 3954658 and 3971212 may **only** be mixed with each other.
- Connecting rod Part Number 4891176 **must never** be mixed with any other connecting rod.
- Connecting rod Part Number 3942581 **must never** be mixed with any other connecting rod.